ECGR 4101/5101

Requirements of Lab 8: Tic Tac Toe Game

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Microcontroller Requirements

Req_MC001: Each player module should have one MSP430G2533 microcontroller.

Req_MC002: The microcontroller must be in a 20-pin DIP package.

Req_MC003: The microcontroller should control the game logic.

Req_MC004: The microcontroller should control the sequence of player turns.

Req_MC005: The microcontroller should validate proper movement between player modules.

Power Supply Requirements

Req_Pow001: Each player module must have its own independent power source (battery).

Req_Pow002: The system must utilize a voltage regulator to ensure a consistent 3.3V output for each MSP430G2533 microcontroller.

Req_Pow003: The power supply must sustain a 3.3V output with a tolerance of ±5% under all game operating conditions.

Req_Pow004: The power supply should provide sufficient current to operate both MSP430G2533 microcontrollers and all connected peripherals, including the LCD, LEDs, and buzzer.

User Interface Requirements

Req_UI001: The system must have an LCD to display the Tic Tac Toe grid game.

Req_UI002: The LCD should display messages such as “Start,” “Game Over,” “Player A’s turn”, “Player B’s turn”, “Player A Wins,” and “Player B Wins.”

Req_UI003: The LCD must indicate the current player’s turn.

Req_UI004: Each player should have a button to navigate across the Tic Tac Toe grid.

Req_UI005: Each player should have a button to place their selected mark (X or O) on the chosen grid position.

Audio and Visual Requirements

Req_AV001: Each player module must include an LED.

Req_AV002: The LED must blink to indicate the player's turn.

Req_AV003: Each player module must include a buzzer to signal when a player places their move.

Req_AV004: The buzzer must emit a distinct sound each time the player navigates to a new grid position.

Req_AV005: The buzzer must emit a different, distinct sound when the player places their mark.

Req_AV006: The LCD display for the opposing player must show only the final placement of the other player's mark (X or O) without displaying the intermediate navigation across the grid.

Game Operation Requirements (Newly Added to Avoid Overlap)

Req_Game001: The game must allow either of the players to start a new game once the previous game is completed.

Req_Game002: The system must allow each player to choose whether they want to represent "X" or "O" at the start of the game.

Req_Game003: During gameplay, the system must ensure that the player's move is visible only on their own board and not directly visible to the opposing player.

Game Communication Requirements

Req_Comm001: The two player modules must communicate through a cabled connection.

Req_Comm002: The communication cable should be at least 3 feet in length.

Req_Comm003: The communication link should be transmitted and receive data without interference.

Hardware Layout Requirements

Req_HW001: All components should be assembled on a custom PCB (Printed Circuit Board).

Req_HW002: The PCB must not use any jumper wires.

Req_HW003: Headers should be included to enable in-circuit programming of the microcontroller.

Testing Requirements

Req_Test001: Each button should be tested individually to ensure accurate functionality and responsiveness.

Req_Test002: The LED should blink exclusively during the respective player's turn.

Req_Test003: The buzzer must sound only upon confirmation of movement.

Req_Test004: The I2C communication link among the two microcontrollers and the LCD should be tested to ensure proper data communication and accurate display functionality.

Req_Test005: The game logic must be tested to ensure proper handling of player turns, move validation, win conditions, and draw scenarios.

Operational Reliability Requirements

Req_Relia001: The system must ensure all button inputs are processed without any lag.

Req_Relia002: The LCD must refresh the displayed board's state immediately after a move is placed.

Programming and Code Requirements

Req_Prog001: The program must be written entirely in the C programming language.

Req_Prog002: The program should not use inbuilt delay cycles for timing adjustments.

Req_Prog003: The program must be structured, formatted, and readable based on general coding guidelines.

Req_Prog004: The program must use functions wherever possible to ensure modularity and reusability.

Project Design and Assembly Constraints

Req_Const001: The team must adhere to the deadlines for Billing of Materials, schematic, test plan, and demo.

Req_Const002: The design must use MSP430G2533 microcontrollers only.

Req_Const003: The assembly should use a soldered PCB.